

Divide Fractions

Lesson 2-3

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

12 dots

$$\frac{12}{3} = 4$$

In $\frac{3}{4} \div \frac{1}{2}$, the dividend is $\frac{3}{4}$ — the total amount being split

Dividend

The number you are splitting up in a division problem.



$$12 \div 3 = 4$$

In $\frac{3}{4} \div \frac{1}{2}$, the divisor is $\frac{1}{2}$ — the size of each group

Divisor

The number you split by in a division problem.

$$\frac{3}{4} \rightarrow \frac{4}{3}$$

flip top and bottom

The reciprocal of $\frac{2}{3}$ is $\frac{3}{2}$. To divide by $\frac{2}{3}$, multiply by $\frac{3}{2}$.

Reciprocal

A fraction turned upside down.

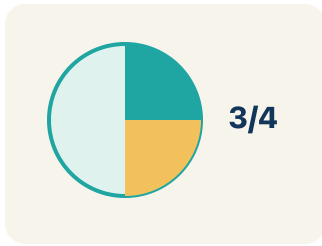


$$12 \div 3 = 4$$

In $\frac{3}{4} \div \frac{1}{4} = 3$, the quotient is 3 — three $\frac{1}{4}$ -size pieces fit into $\frac{3}{4}$

Quotient

The answer when you divide.



$$8/12 \rightarrow \text{GCF is } 4 \rightarrow 8 \div 4 / 12 \div 4 = 2/3$$

Simplify

To make a fraction smaller using the same parts,
like $2/4 = 1/2$.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can divide a fraction by a fraction by multiplying by the reciprocal and simplifying.

1 Notice

Agent Rivera is dividing a $\frac{3}{4}$ -pound evidence bag into portions that each weigh $\frac{1}{8}$ pound.

2 Key idea

I can divide a fraction by a fraction by multiplying by the reciprocal and simplifying.

3 Words I'll use

Dividend

Divisor

Reciprocal

Quotient

Simplify



Think About It

- What is the total weight being divided?
- What is the size of each portion?
- Will the answer be more or less than 1?

See the notes in action: watch one worked all the way through, then try the next with the same steps.



I do — watch

Follow each step as your teacher solves it.

Problem: What is the reciprocal of $\frac{3}{5}$?

A. $\frac{5}{3}$

B. $\frac{3}{5}$

C. $\frac{5}{5}$

D. $\frac{1}{3}$

Step 1

To find the reciprocal, flip the fraction.

Step 2

The reciprocal of $\frac{3}{5}$ is $\frac{5}{3}$.



Answer: A. $\frac{5}{3}$

 We do — together

Solve this one with your class using the same steps.

Problem: What is $\frac{1}{2} \div \frac{1}{4}$?

- A. 2
- B. $\frac{1}{8}$
- C. 4
- D. $\frac{1}{2}$

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: What is $\frac{3}{4} \div \frac{1}{8}$?

- A. 6
- B. $\frac{3}{32}$
- C. $\frac{8}{3}$
- D. 24

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Agent Rivera splits a $\frac{3}{4}$ -pound evidence bag into $\frac{1}{8}$ -pound portions. Will she get more or fewer than 1 portion? Estimate before solving.

Level 1 support

Start here: She will get many portions because $\frac{1}{8}$ pound is much smaller than $\frac{3}{4}$ pound.

 **Need a hint?**

Sentence starters (optional)

She will get ____ than 1 portion because ____.

Ella obtendrá ____ que 1 porción porque ____.

Since $\frac{3}{4}$ is the dividend and $\frac{1}{8}$ is the divisor, ____.

Como $\frac{3}{4}$ es el dividendo y $\frac{1}{8}$ es el divisor, ____.

Word bank: dividend divisor quotient

Listen for: Students recognize the divisor $\frac{1}{8}$ is smaller than the dividend $\frac{3}{4}$, so the quotient is greater than 1.

Level 2

Estimate the exact number of portions and explain your reasoning using equivalent fractions before computing.

- I estimate about ____ portions because $\frac{3}{4}$ equals ____ eighths.
- Comparing eighths to eighths shows ____.

EXPLORE

Why do we multiply by the reciprocal when we divide one fraction by another?

Level 1 support

Start here: Multiplying by the reciprocal undoes the divisor and gives the same result as dividing.



Need a hint?

Sentence starters (optional)

I flipped the divisor to its reciprocal because ____.

Volteé el divisor a su recíproco porque ____.

After Keep, Change, Flip the answer is ____.

Después de Mantener, Cambiar, Voltear la respuesta es ____.

Word bank: reciprocal divisor quotient simplify

Listen for: Students explain that flipping the divisor turns division into multiplication, and they simplify the final quotient.

Level 2

Show why $\frac{3}{4} \div \frac{1}{8}$ gives the same answer whether you count eighths in $\frac{3}{4}$ or multiply by the reciprocal.

- Both methods agree because ____.
- Counting eighths matches multiplying by 8 because ____.

CONNECT

How could dividing fractions help you share food or materials fairly?

Level 1 support

Start here: Dividing fractions tells you how many equal fraction-size shares fit into an amount.



Need a hint?

Sentence starters (optional)

Dividing fractions helps when ____.

Dividir fracciones ayuda cuando ____.

For example, ____ of a ____ split into ____ shares gives ____.

Por ejemplo, ____ de un ____ dividido en ____ partes da ____.

Word bank: **dividend** **divisor** **simplify**

Listen for: Students give a fair-share scenario and identify which fraction is the dividend and which is the divisor.

Level 2

A classmate flipped the dividend instead of the divisor. Critique this error and explain the correct step.

- Flipping the dividend is wrong because ____.
- Only the ____ should be flipped because ____.

Write About the Math

The Writing Revolution

I can explain my steps using the words dividend, divisor, reciprocal, quotient, and simplify.

1. Kernel Sentence subject + verb

Model: Simplify is to make a fraction smaller using the same parts, like $2/4 = 1/2$.

Simplificar es hacer una fracción más pequeña con las mismas partes, como $2/4 = 1/2$.

Write a kernel sentence about simplify. Use a subject and a verb.

Escribe una oración base sobre simplificar. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Simplify matters in math

Simplificar importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Simplify matters in math because ____.

Simplificar importa en matemáticas porque ____.

but
pero

Simplify matters in math, but ____.

Simplificar importa en matemáticas, pero ____.

so
entonces

Simplify matters in math, so ____.

Simplificar importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about simplify.
Di un hecho verdadero sobre simplify.

Simplify ____.

Question *Pregunta*

Ask a question about simplify.
Haz una pregunta sobre simplify.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about simplify.
Muestra entusiasmo sobre simplify.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with simplify.
Dile a un compañero qué hacer con simplify.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

She will get ____ than 1 portion because ____.

Ella obtendrá ____ que 1 porción porque ____.

Since $\frac{3}{4}$ is the dividend and $\frac{1}{8}$ is the divisor, ____.

Como $\frac{3}{4}$ es el dividendo y $\frac{1}{8}$ es el divisor, ____.

I flipped the divisor to its reciprocal because ____.

Volteé el divisor a su recíproco porque ____.

Try It

Solve on your own. Check the answer key when you are done.

1. What is $(5/6) \div (5/12)$?

- A. 2
- B. $1/2$
- C. $25/72$
- D. $12/5$

Show your work:

2. What is $(7/8) \div (1/4)$?

- A. $7/2$
- B. $7/32$
- C. $1/2$
- D. $4/7$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Is $\frac{3}{4} \div \frac{1}{2}$ the same as $\frac{1}{2} \div \frac{3}{4}$? Solve both and explain why or why not. What does this tell you about the order in division?

Sentence starter: $\frac{3}{4} \div \frac{1}{2} = \underline{\hspace{1cm}}$ and $\frac{1}{2} \div \frac{3}{4} = \underline{\hspace{1cm}}$. These are $\underline{\hspace{1cm}}$ because $\underline{\hspace{1cm}}$. This shows that division is $\underline{\hspace{1cm}}$.

Show your work:

Reflect — Exit Ticket

What is $\frac{4}{5} \div \frac{2}{5}$?

- A. 2
- B. $\frac{8}{25}$
- C. $\frac{2}{5}$
- D. $\frac{10}{5}$

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. $2 - 1/2 \div 1/4 = 1/2 \times 4/1 = 4/2 = 2$. *Two 1/4-size pieces fit into 1/2.*
2. **You Try (notes):** A. $6 - 3/4 \div 1/8 = 3/4 \times 8/1 = 24/4 = 6$. *Six 1/8-size pieces fit into 3/4.*
3. **Try It 1:** A. $2 - (5/6) \times (12/5) = 60/30 = 2$.
4. **Try It 2:** A. $7/2 - (7/8) \times (4/1) = 28/8 = 7/2$.
5. **Exit Ticket:** A. $2 - 4/5 \div 2/5 = 4/5 \times 5/2 = 20/10 = 2$. *Two groups of 2/5 fit into 4/5.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Simplify is to make a fraction smaller using the same parts, like $2/4 = 1/2$.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).